

Frontiers in Neurorobotics

Scope drift deep-dive & editorial options

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Sophie Wilson · Advanced Analytics · Further work

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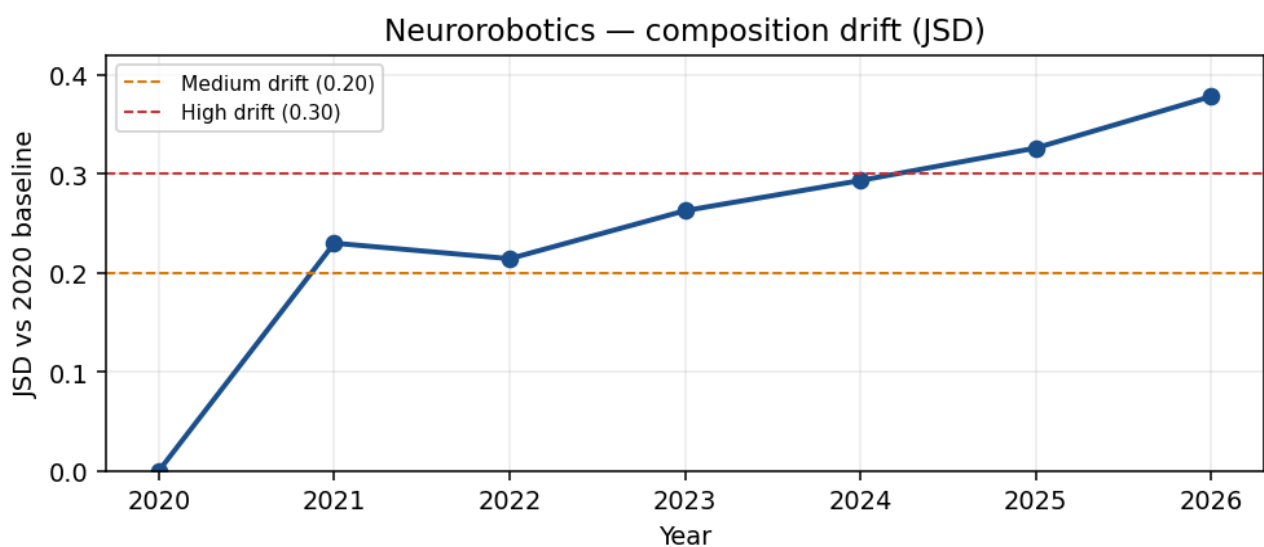
1. Executive summary

Against a **2020 baseline**, Neurorobotics shows clear composition drift from 2021, driven mainly by a shift toward **computer vision** and away from **neuroscience**, not by a collapse of the primary community set. Out-of-scope rate stays roughly flat (~16–25%). A meaningful minority of “OOS” papers look like **false OOS** (neurorobotics titles parked in far-field Leiden clusters).

Onset 2021 JSD ≥ 0.20	JSD 2026 0.38 vs 2020	OOS 2026 16.7% 30 papers	Primary set Unchanged same 4 clusters	All-years OOS 20.8% 1,037 papers
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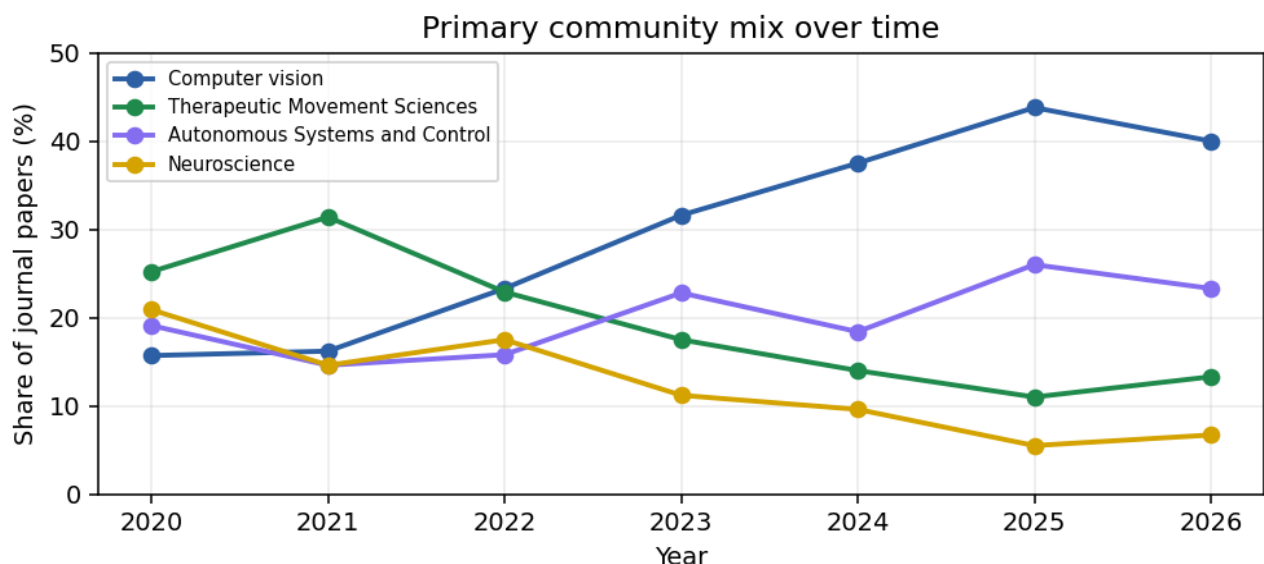
2. When did drift start?

JSD jumps in **2021** ($0 \rightarrow \sim 0.23$), then climbs through 2023–2025 to **0.38** in 2026. Treat 2021 as onset; weight 2023–25 more than early 2026 (small n).



3. How is the journal drifting?

Primary community **IDs are unchanged** (Computer vision, Therapeutic Movement Sciences, Autonomous Systems & Control, Neuroscience), but **shares flipped**: vision rose from ~16% (2020) to ~40%+ (2025–26); neuroscience fell from ~21% to ~6–7%.



Year	n	OOS%	CV%	TMS%	Auto%	Neuro%
2020	115	19.1	15.7	25.2	19.1	20.9
2021	185	23.2	16.2	31.4	14.6	14.6
2022	292	21.6	23.3	22.9	15.8	17.5
2023	206	18.0	31.6	17.5	22.8	11.2
2024	136	25.0	37.5	14.0	18.4	9.6
2025	73	16.4	43.8	11.0	26.0	5.5
2026	30	16.7	40.0	13.3	23.3	6.7

Primary shift flag (2020→2026): **unchanged set**. Gain/loss labels: none — drift is reweighting, not replacement of the core set.

4. Is OOS genuine?

Mostly yes for far-field clusters; partly no for misplaced neurorobotics. LLM borderline labelled all non-primary candidates as hard OOS (materials, IoT security, geology, thermal, etc.). That is appropriate for true far-field work. However, titles in Materials / Business / Language–Health clusters often still read as robot–neural–rehab — these are **false or soft OOS** from Leiden macro assignment.

4.1 Contested OOS examples (keyword overlap)

Sample of community-OOS titles that mention robot / neural / prosthetic / embodied terms (truncated). These are candidates for expand/keep, not automatic reject:

- 2020 · C0 — Examples of Gibsonian Affordances in Legged Roboti
- 2020 · C0 — Editorial: Tactile Intelligence in Robots
- 2020 · C0 — Surgeon-Centered Analysis of Robot-Assisted Needle
- 2020 · C18 — The DIAMOND Model: Deep Recurrent Neural Networks
- 2020 · C20 — A Custom EOG-Based HMI Using Neural Network Modeli
- 2020 · C22 — SAUV—A Bio-Inspired Soft-Robotic Autonomous Underw
- 2021 · C0 — A Neural Network Model for Learning 3D Object Repr
- 2021 · C1 — Prosthetic Embodiment and Body Image Changes in Pa
- 2021 · C3 — Editorial: Integrated Multi-modal and Sensorimotor
- 2021 · C3 — Sensor-Based Control for Collaborative Robots: Fun
- 2021 · C18 — Neurorobotic Models of Neurological Disorders: A M
- 2021 · C18 — Robot Communication: Network Traffic Classificatio

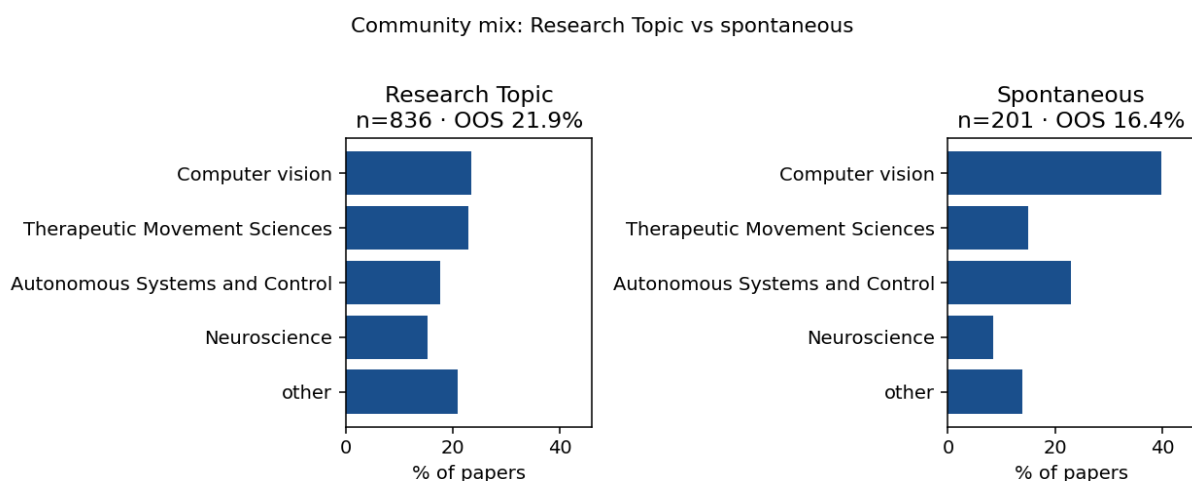
Genuine OOS to keep rejecting: thermal energy, geology, cancer immunotherapy, organic chemistry, urban mobility, generic management without robots, pure remote-sensing / medical X-ray CV without embodiment.

5. Research Topics — OOS, drift, keep/cut

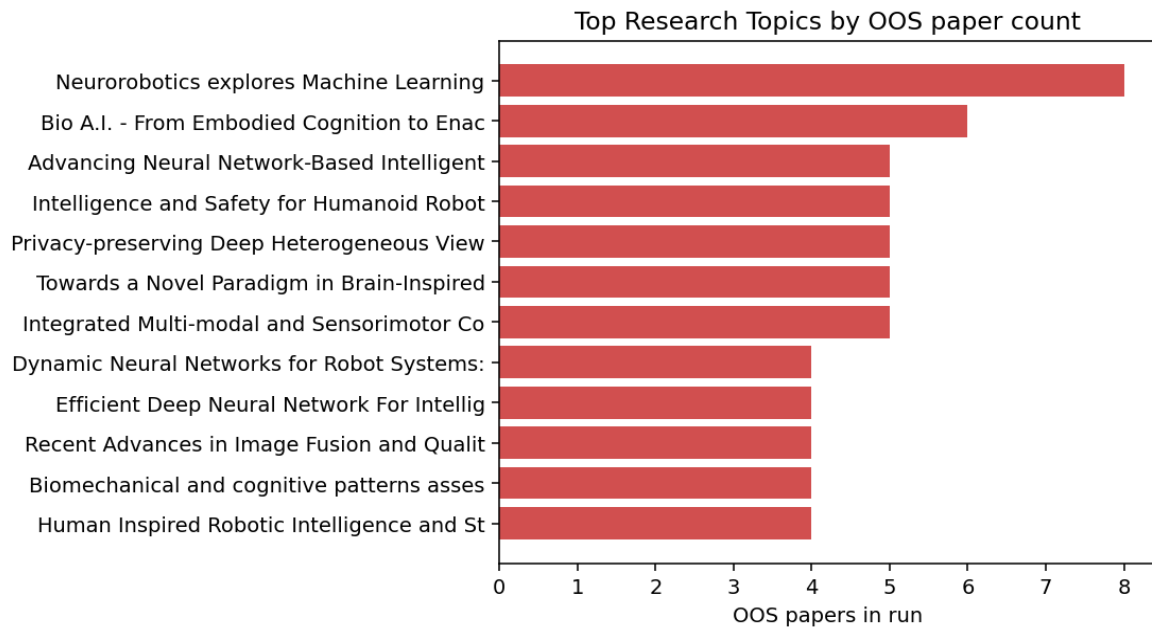
Spontaneous submissions are more CV-heavy than RTs (~40% vs ~23% CV) and slightly less OOS (~16% vs ~22%). So the CV takeover is not 'RTs alone' — open submissions drive mix shift; RTs add OOS risk and volume, especially ML/vision/cyber-physical topics (2020-22 launch wave). Close/gate off-brand high-OOS RTs; add rehab / neural-interface / embodied-vision RTs to rebalance identity.

5.1 RT vs spontaneous submissions

Of 1037 run papers, **836** are in a Research Topic and **201** are spontaneous. OOS: RT **21.9%** vs spontaneous **16.4%**. Computer vision share: RT **23.4%** vs spontaneous **39.8%**. Neuroscience share: RT **15.2%** vs spontaneous **8.5%**. Spontaneous is the stronger CV-mix driver; RTs carry more OOS risk.



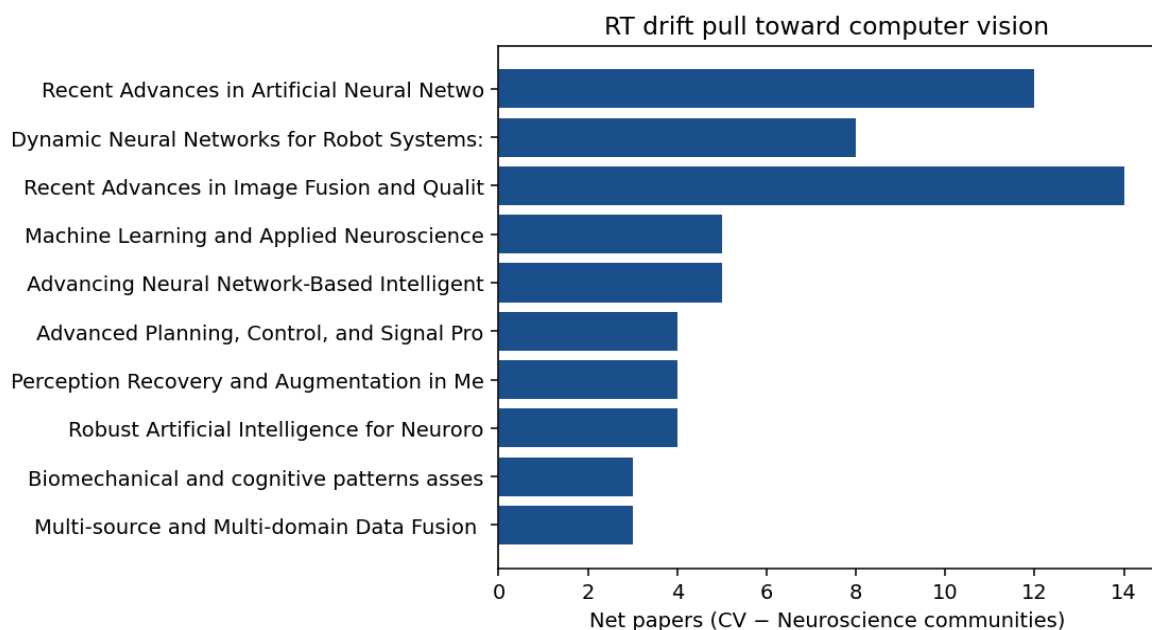
5.2 Which RTs contribute to OOS?



RT (short)	n	OOS%	Action
Neurorobotics explores Machine Learning	17	47.1%	gate (robot/embodied require
Bio A.I. - From Embodied Cognition to Enactive R	19	31.6%	review / tighten scope
Advancing Neural Network-Based Intelligent Algor	16	31.2%	gate (robot/embodied require
Intelligence and Safety for Humanoid Robots: Des	11	45.5%	review / tighten scope
Privacy-preserving Deep Heterogeneous View Perce	8	62.5%	remove / do not renew
Towards a Novel Paradigm in Brain-Inspired Compu	8	62.5%	gate (robot/embodied require
Integrated Multi-modal and Sensorimotor Coordina	14	35.7%	review / tighten scope
Dynamic Neural Networks for Robot Systems: Data-	21	19.0%	keep

5.3 Which RTs contribute to drift?

Drift pull $\approx$ papers in the Computer Vision community minus Neuroscience for each RT. Image-fusion / ANN-embedded-systems series are pure CV volume with little neural-robot content — they move the mix even when OOS% is moderate.



5.4 Remove / gate / add

Heuristic on 150 RTs with papers in the run: **11** remove/do-not-renew or tight-gate; **16** gate with robot/emodied requirement; **60** keep/grow. Brand series (Women/Horizons/Insights) kept for audit, not auto-cut.

Remove / do not renew (priority):

- **Privacy-preserving Deep Heterogeneous View Perception for Data Learnin** — n=8, OOS 62.5% · remove / do not renew
- **Recent Advances in Image Fusion and Quality Improvement for Cyber-Phys** — n=8, OOS 50.0% · remove / do not renew
- **Biomechanical and cognitive patterns assessment in human-machine colla** — n=7, OOS 57.1% · remove / tight gate
- **Human Inspired Robotic Intelligence and Structure in Demanding Environ** — n=7, OOS 57.1% · remove / tight gate
- **Towards Neuroscience-Inspired Intelligent Computing: Theory, Methods,** — n=4, OOS 100.0% · remove / tight gate
- **Human-in-the-loop Cyber-Physical Systems** — n=6, OOS 50.0% · remove / do not renew
- **Robotics, Neural Networks and Cyber-Physical Systems Applied to Sustai** — n=7, OOS 42.9% · remove / do not renew
- **Deformable Object Manipulation** — n=4, OOS 75.0% · remove / tight gate
- **Recent Advances in Image Fusion and Quality Improvement for Cyber-Phys** — n=14, OOS 7.1% · remove / do not renew
- **Recent Advances in Artificial Neural Networks and Embedded Systems for** — n=13, OOS 7.7% · remove / do not renew

Add (proposed new RTs):

- **Embodied robot vision & multimodal perception** — CV already dominates volume; formalise robot-gated vision so spontaneous/CV RTs don't stretch OOS.
- **Therapeutic & assistive robotics (exoskeletons, prostheses, rehab)** — Therapeutic Movement share fell; dedicated RT can rebuild neural-rehab identity.
- **Neural interfaces ↔ robot control (BCI, motor decoding, neurorobotic loops)** — Neuroscience share collapsed to ~6–7%; RT can re-anchor the journal name.
- **Learning & control for physical agents (embodied RL, adaptive control)** — Aligns with Autonomous Systems primary; keeps ML on-robot rather than generic ML.

Full table: `further_work/neurorobotics_rt_deep_dive.csv` (re-run `python further_work/analyze_rts.py`).

6. Editorial options

6.1 Expand scope?

Yes, selectively. Formalise what you already publish: embodied / robot vision & learning; therapeutic & assistive robotics; neural interfaces ↔ robot control. **Do not** expand into materials, IoT cybersecurity, traffic forecasting, or generic NLP.

6.2 Rename?

Strong case. “Neurorobotics” undersells vision/control dominance and over-promises neuroscience (~6–7% recently). Options: *Neurorobotics and Embodied AI*; *Neural and Robotic Systems*; or keep the name with an explicit scope line covering neural interfaces, embodied intelligence, and robot-assisted motor systems.

6.3 Did a section launch cause the drift?

No. Taxonomy join shows all papers as **Specialty Journal** with null section — no Specialty Section rows. Use Research Topics (§5) for launch-vs-drift attribution instead.

6.4 New sections — where (forward architecture)?

1) Neural interfaces & motor neuroscience · 2) Therapeutic & assistive robotics · 3) Embodied vision & multimodal perception (robot-centric gate) · 4) Learning & control for physical agents · 5) Optional social/assistive HRI only if you choose to expand. Gate sections 3–4 with “must involve a robot / embodied agent / neural–motor loop”. These are **recommended future sections**, not an explanation of past drift.

7. Recommended next steps

- Treat 2021 as onset; use 2023–25 as the CV-takeover window in JD narratives.
- Act on RT list: close off-brand / high-OOS topics; gate ML–vision RTs with robot/embodied requirement.
- Do not attribute drift to section launches — none exist.
- Manually audit false OOS in clusters 0 / 3 / 20 (Materials, Business, Lang–Health).
- Choose strategy A (broaden + rename toward embodied AI) or B (tighten to neural–robot + rehab and desk-reject pure CV).
- Stand up sections / on-brand RTs under the chosen strategy; do not add materials/IoT topics.
- Re-check mid/late 2026 volume before locking 2026-only conclusions.

Disclaimer: This brief was created with AI assistance from dashboard outputs of the scope-drift pipeline. Community labels are model-generated; contested OOS needs editorial review.